GASOLINE-ENGINE CALCULATION — MAIN DIAGRAMS

Toyota 1ZR-FAE: geometric baseline · 4 400 min⁻¹ · 1,598 L · 59,42 kW · 129 N·m

ICE 0.2.0
A1 · 841 × 594 mm

1 / INDICATOR DIAGRAM

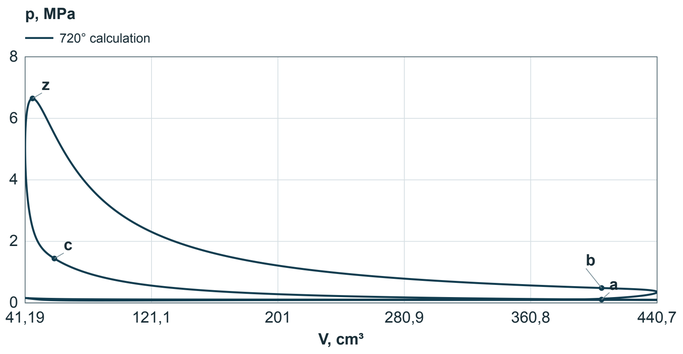


Figure 7.1 — Full-cycle indicator diagram

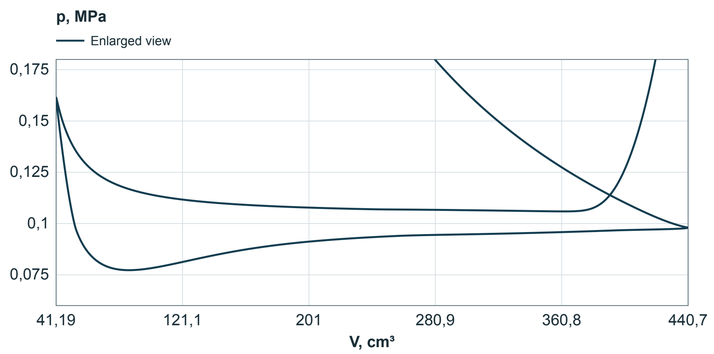


Figure 7.2 — Low-pressure p–V region

2 / SPEED CHARACTERISTIC

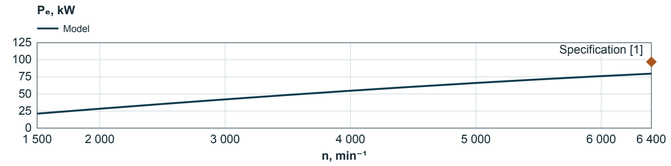


Figure 9.1 — Model power and rated point

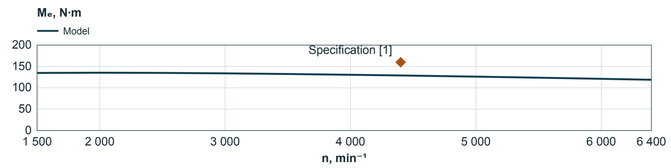


Figure 9.2 — Model torque and rated point

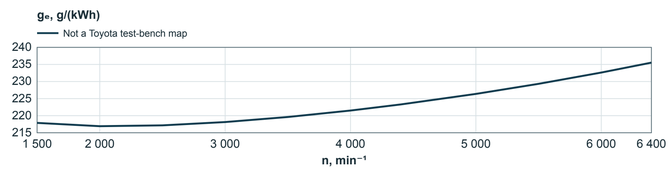


Figure 9.3 — Calculated specific fuel consumption

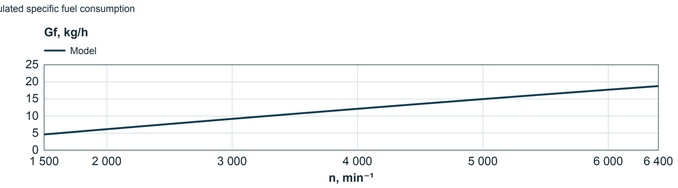


Figure 9.4 — Calculated hourly fuel consumption

3 / PISTON KINEMATICS

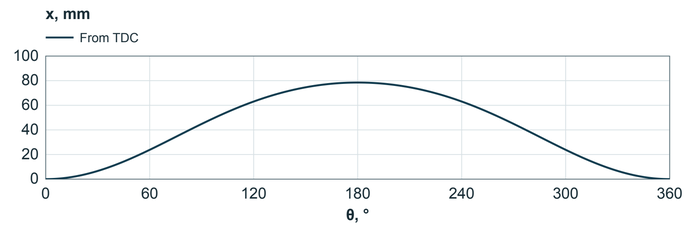


Figure 10.1 — Piston displacement

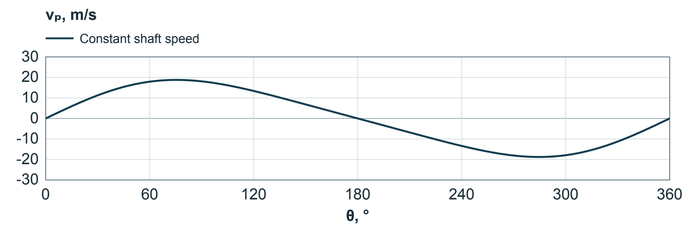


Figure 10.2 — Piston speed

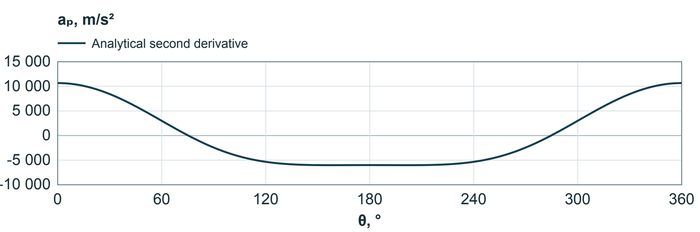


Figure 10.3 — Piston acceleration

4 / AXIAL LOADS

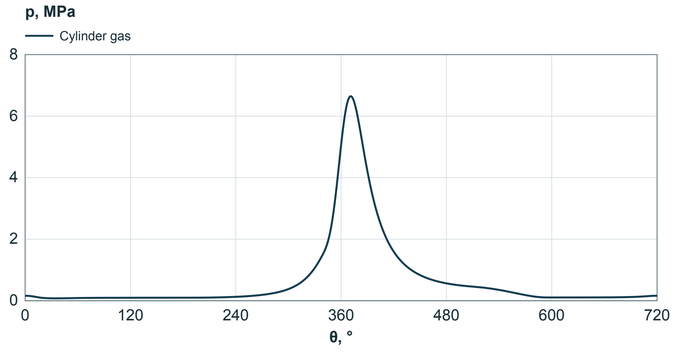


Figure 5.1 — Pressure versus crank angle

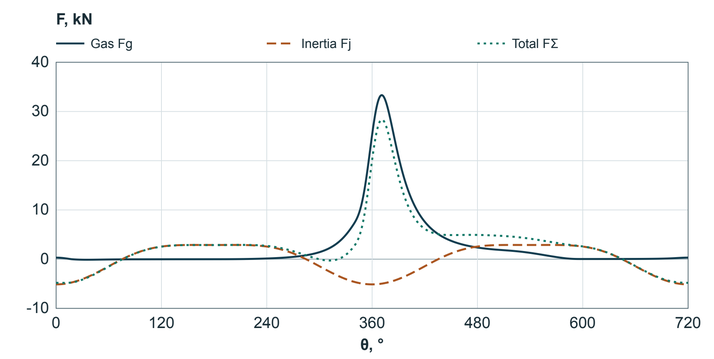


Figure 11.1 — Single-cylinder axial forces

5 / CONNECTING ROD AND CRANK

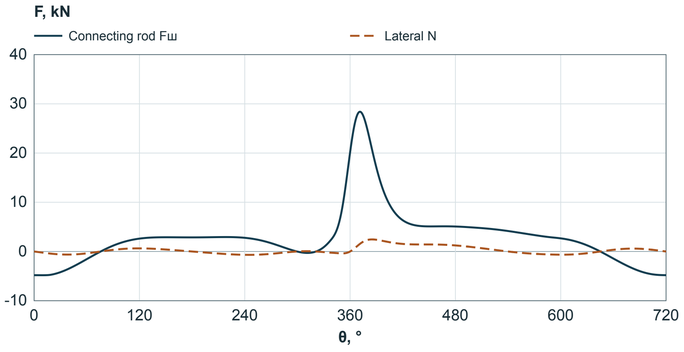


Figure 11.2 — Connecting-rod force and side reaction

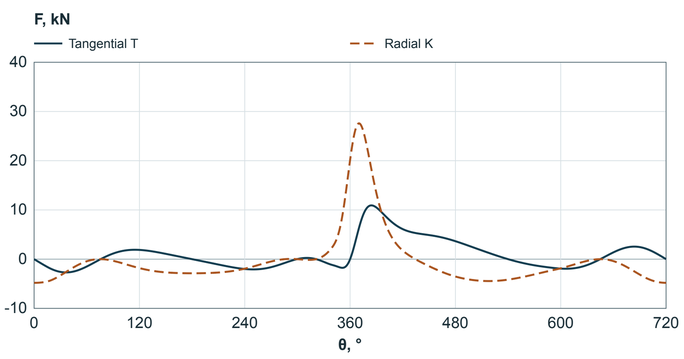


Figure 11.3 — Components of force transmitted by the connecting rod

6 / TORQUE

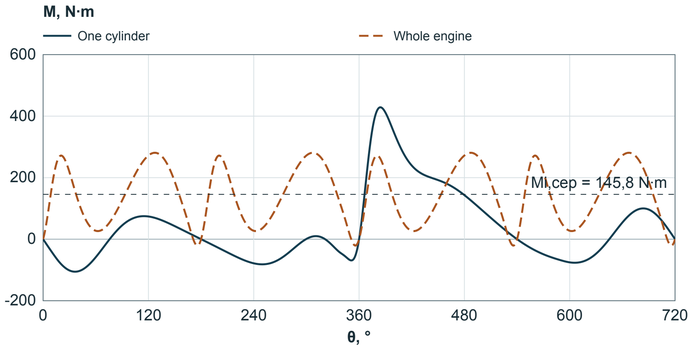


Figure 11.4 — Single-cylinder and whole-engine torque

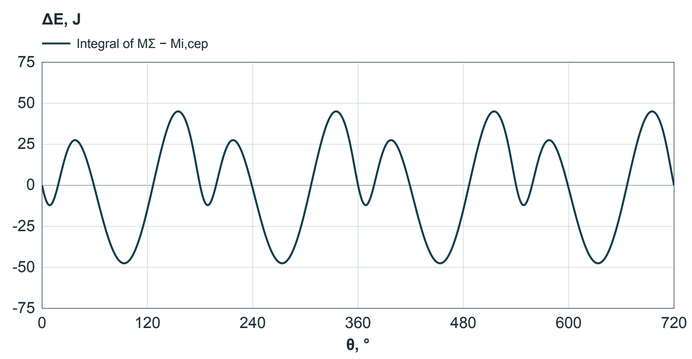


Figure 11.5 — Excess energy at constant speed

Uncalibrated single-zone model. Geometry [2]; rated points [1]. Timing, masses and losses are assumptions.
Figure and source numbers match the A4 report. This is a diagram appendix, not an assembly drawing or proof of Euro 6 compliance.

θ = 0° — gas-exchange TDC; θ = 360° — combustion TDC.
Every axis has its own units. Mean torque line is P/ω.